UNIVERSITY OF WOLLONGONG AUSTRALIA

Tutorial 10 — Optional

P, PI and PID: design and implementation checks

Optional controller-design study after the eight-class core course, separate from Project 2.
Optional independent practice: start with Q1, then Q2–Q4. Sections labelled Extension or After class are independent practice. Show reasoning, substitutions and units.

Question 1 Quick checks on controller actions and integral recovery

Give a brief reason or calculation.

a) Identify controller actions. Classify $C_1(s)=2$, $C_2(s)=2+0.1/s$ and $C_3(s)=2+0.1/s+0.5s$. Which term accumulates error and which responds to its rate?

b) A proportional command. A temperature controller has $K_p=2$ W/°C, error +0.4°C and no other active controller term. Find its requested power. Does a nonzero constant error make this power grow with time?

c) Differentiate the signal. With $K_d=0.5$ W·s/°C, find ideal derivative contribution for constant error +0.4°C and for error increasing at 2°C/s. Why does a reference step cause difficulty when differentiating error?

d) Conditional integration. A heater has limits 0–5 W and $K_i=0.036$ W/(°C·s). For upper saturation with error +0.4 or −0.4°C, and lower saturation with error −0.4 or +0.4°C, state whether to block or allow integration and find its rate. Does an 8 W integral contribution alone give delivered power?

Question 2 A PI temperature controller with a finite heater

The chamber temperature-rise plant below has gain in °C/W. The heater can deliver 0–5 W. The reference is a 12°C rise from a zero initial state. First use P gain K_p = 0.5 W/°C. Then design PI control for target poles −0.04 and −0.06 s^{-1}. Assume unity negative feedback; distinguish linear predictions from the limited actuator.

$$G(s)=\frac{3}{45s+1},\ C_{PI}(s)=K_p+\frac{K_i}{s}$$

a) For P control, find closed-loop transfer function, final temperature rise, steady error and initial power demand. Is the target physically attainable in steady state?

b) For PI control, match the polynomial and find both gains and units. Write the actual closed-loop transfer function with its numerator.

c) Check PI initial and steady power demands. After class: invert the unsaturated step response and check its initial value, initial slope and final value. Explain what clipping invalidates.
MATLAB verification. After designing the PI controller, plot the unsaturated temperature response and requested power. Simulate the 0–5 W heater with conditional integration: block integration when the total request reaches a limit and the error would worsen saturation.

Submit one figure with temperature and power in separate panels, overlaying the linear and limited responses. Use contrasting colours and line styles. Report and explain their initial slopes. Base MATLAB is sufficient.

Question 3 Audit two proposed PID settings

A separate normalized second-order process and ideal parallel PID are given below. An additive disturbance D enters at the plant input after the controller. Use unity negative feedback. Desired characteristic polynomial is $(s + 8)(s^2 + 6s + 25)$. Candidate A has (K_d, K_p, K_i) = (10, 57, 200); candidate B has (14, 73, 200).

$$G(s)=\frac{1}{s^2+4s+16},\ C(s)=K_d s+K_p+\frac{K_i}{s}$$

a) Expand the desired polynomial and derive the actual coefficient-matching equations. Decide which candidate is correct and diagnose the other.

b) For the correct candidate, find Y/R and Y/D and identify the poles. Does a step input disturbance leave a steady output offset?

c) Explain why a denominator match does not justify a zero-free second-order overshoot formula. What is the practical problem with ideal differentiation of a reference step?

Question 4 A derivative filter changes noise and poles

After class
Keep the correct PID gains of Question 3, but replace the ideal derivative with the filtered branch below. A noise sinusoid of amplitude 0.003 normalized output units at 100 rad/s is applied to that branch. Compare T_f = 0.05 s and 0.10 s. This branch calculation is not the total closed-loop noise response.

$$C_D(s)=\frac{K_d s}{1+T_f s}$$

a) Calculate derivative-branch command amplitude for both filters and for an ideal derivative. Give the high-frequency gain limit.

b) Derive the complete filtered-PID characteristic polynomial. Calculate its coefficients for both T_f values and check the poles numerically.

c) What must be rechecked before treating this filtered implementation as the original target design?

d) For a unit step applied directly to the derivative input, find the filtered branch command and its initial peak for both filters. How does differentiating measured output avoid reference kick?

Method reference

Quick-check sources: Ogata, System Dynamics, 4th ed., Sec. 10–3, pp. 501–505 (controller actions); Modern Control Engineering, 5th ed., Sec. 8–2, p. 568, and Sec. 8–5, pp. 590–592. The conditional-integration check is independently developed and retained from the previous solution.
Method source: Ogata, Modern Control Engineering, 5th ed., Problem A-8-5, pp. 617–619 (PID design). The thermal PI design, candidate audit and conditional-integration tests below are independently adapted. Page numbers refer to printed pages.
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